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MILITARY SPECIFICATION

NICKEL-TITANIUM ALLOYS, LOW MAGNETIC EFFECTS; BARS, PLATES, SHEETS, STRIPS, AND SHAPES (FOR SPECIAL PURPOSE TOOLS AND EQUIPMENT)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope. This specification covers three different nickel-titanium alloys having low magnetic permeability and having the capability of developing moderate to high mechanical properties by metallurgical processing. (See 6.1.)

1.2 Classification. The nickel-titanium alloys covered by this specification shall be furnished in the following nominal compositions forms, and conditions, or combination of conditions, as specified (see 6.2).

Compositions:

1--55% Ni - 45% Ti
2--56.5% Ni - 43.5% Ti
3--60% Ni - 40% Ti

Forms:

Bar
Plate
Sheet
Strip
Shape

Conditions:

A--Cast
B--Hot worked
C--Cold worked

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D--Annealed
E--Quench hardened
F--Quench hardened and tempered

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

*	MIL-M-19595	Magnetic Effect Limits for Nonmagnetic Equipment Used in the Proximity of Magnetic Influence Ordnance

STANDARDS

Federal

FED-STD-48	Tolerances for Steel and Iron Wrought Products
FED-STD-151	Metals; Test Methods

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-163	Steel Mill Products, Preparation for Shipment and Storage

(Copies of standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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American Society for Testing and Materials

ASTM E 18	Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
ASTM E 140	Standard Hardness Conversion Tables for Metals

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

* 3.1 Chemical composition. The chemical composition of the alloys covered in this specification shall conform to the requirements of table I.

Table I

CHEMICAL COMPOSITIONS (PERCENT)

Element	Composition number		
	1	2	3
Titanium	44.5 - 45.4	43 - 44	39 - 41
Iron, maximum	0.10	0.10	0.10
Other, maximum	0.1	0.1	0.1
Nickel	Balance	Balance	Balance

3.1.1 The contractor shall furnish an analysis of each lot showing the percent by weight of each of the elements specified in table I.

3.2 Hardness. Unless otherwise specified in the contract or order, the conditions (see 1.2) of the forms for the specified alloy compositions shall conform to the Rockwell hardness requirements of table II.

* 3.3 Magnetic effect. All forms of these alloys shall conform to the magnetic effect requirement of paragraph 3.2 in MIL-M-19595.

3.4 Microstructure. The microstructure of the forms of the specified alloy composition shall show a homogeneous distribution of the phases and, for all wrought forms, uniformly sized and equiaxed grains.

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Table II

ROCKWELL HARDNESS REQUIREMENTS

Condition	Composition number		
	1	2	3
Cast	-[1]	-[1]	55 HRC, maximum
Hot worked[2]	25 HRC, maximum	45 HRC, maximum	55 HRC, maximum
Cold worked	-[3]	-[3]	-[3]
Annealed[2]	-[1]	35 HRC, maximum	42 HRC, maximum
Quench hardened[2]	-[1]	40 - 47 HRC	57 - 62 HRC
Quench hardened and tempered[2]	-[1]	30 - 47 HRC	30 - 62 HRC

[1]Hardness is "not required."
[2]See 6.3 for recommended temperature.
[3]Hardness shall be "as required" by the procuring activity but shall not exceed 40 HRC.

3.5 Macrostructure. The macrostructure of the forms of the specified alloy composition shall be dense, sound, free from pipes, fissures, sponginess, abnormal inclusions or segregations, or unusually numerous pinholes.

3.6 Dimensions and tolerances. Dimensions and tolerances for shapes shall be as specified in the contract or order (see 6.2). For the commercial forms, tolerances shown in the following paragraphs of FED-STD-48 shall be applicable:

Form and dimension	Paragraph
Bars, hot-worked:	
Diameter or distance across flats (round or square)	1a1
Distance across flats (hexagon)	1a2
Width and thickness (flats)	1d3
Length	1d4
Straightness (machine straightened)	1d5
Bars, cold-worked:	
Diameter or distance across flats (round or square)	2d1
Distance across flats (hexagon)	2d2

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Width and thickness (flats)	2d3
Length	1d4
Straightness	2d5
Plates:	
Thickness	4a3
Width and length	4a4
Camber	4a9
Flatness	4d5
Sheets:	
Thickness	12a2
Width	12a3
Length	11a6
Resquared	11a10
Camber	11a12
Flatness	11a13
Strips (for widths not covered by the strip tables, use paragraphs specified above for sheet):	
Thickness	14a1
Crown	14a2
Width	14a3
Length	14a4
Camber	14a5

3.7 Identification. All forms or bundles in a lot shall be readily identifiable by bearing one of the following identifications, as specified in the succeeding paragraphs:

(a) Specification number, composition number, condition and hardness, lot number, manufacturer's name, trademark or symbol, and heat number

(b) Specification number, composition number, and condition and hardness.

3.7.1 Bars and shapes. Bars, three-fourths inch or larger in diameter or across parallel flats, and shapes of equivalent cross-sectional area shall be rubber stamped or marked with a nonwater-soluble ink or die stamped on or near one end with the information specified in 3.7(b).

3.7.2 Bars and shapes. Bars, less than three-fourths inch in diameter or across parallel flats, and shapes of equivalent cross-sectional area shall be bundled and tagged at each end with firmly attached oil-proof or metal tags containing the information specified in 3.7(a).

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3.7.3 Plates, sheets, and flat strips. Plates, sheets, and flat strips shall be identified as specified in 3.7(b) using the stamping or marking procedures of 3.7.1.

3.7.4 Coiled sheets and strips. When sheets and strips are shipped in coils and the size permits, the loose ends of the coils shall be identified as specified in 3.7(a) using the stamping or marking procedure of 3.7.1. On coiled strips too narrow for stamping or marking, identification shall be provided by the procedure of 3.7.2.

3.7.5 Shapes, forged. Each forging whether shipped loose or in bundles shall, whenever possible, carry the identification preferably of 3.7(a) or secondly of 3.7(b) using either the marking or stamping procedure of 3.7.1. When individual marking is not practical for small forgings shipped in bundles, the bundles shall carry the identification of 3.7(a) using the identifying procedure of 3.7.2.

3.8 Workmanship. All forms shall have uniform surface finishes. The surfaces shall be commercially smooth, free from scale, corrosion, cracks, scratches, seams, laps or folds, slivers, and any injurious defect which would limit or impair its serviceability for the intended use.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. A lot shall consist of material of one form, size, condition, and melt, or fractions thereof, processed and manufactured under the same conditions when submitted for inspection and acceptance.

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4.3 Sampling.

4.3.1 For chemical analysis. Sample selection and preparation shall be in accordance with method 111 of FED-STD-151. Sources, sizes, and number of samples shall be in accordance with either one of the two following procedures for the test of 4.5.1:

(a) From the usable portion of each ingot, two individual samples shall be taken; each sample shall contain not less than 2 ounces of millings or drillings.

(b) From each lot of the specified form, one composite sample containing a minimum of 50 grams of clean millings, drillings, or clippings shall be taken. If the lot contains less than four lengths, the composite sample shall be formed from specimens from each length.

* 4.3.2 For hardness testing. From each lot, a total of five specimens from five individual lengths selected at random shall be taken and prepared in accordance with ASTM E 18. If the lot consists of less than five lengths, a specimen shall be taken from each length. Testing shall be in accordance with 4.5.2.

4.3.3 For magnetic effects test. From each lot, a total of five specimens shall be taken from five individual lengths selected at random. If the lot consists of less than five lengths, a specimen shall be taken from each length. Unless otherwise specified in the contract or order (see 6.2), the size of each specimen shall have a minimum volume of three-fourths cubic inch. Testing shall be in accordance with 4.5.3.

4.3.4 For microstructural examination. From each lot a minimum of three metallographic specimens shall be taken of transverse locations, where possible, from three individual lengths selected at random. If the lot consists of less than three lengths, a specimen shall be taken from each length. Testing shall be in accordance with 4.5.4.

4.3.5 For macrostructural examination. From each lot a minimum of three macrostructure specimens shall be taken from three individual lengths selected at random. If the lot consists of less than three lengths, a specimen shall be taken from each length. Each specimen shall be representative of the central cross-sectional area of the specified form and, where possible, shall include the entire cross-sectional area of the related length. Testing shall be in accordance with 4.5.5.

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4.3.6 For visual and dimensional examination.

4.3.6.1 Bars, plates, shapes, and flat sheets and strips. Bars, plates, shapes, and flat sheets and strips shall be sampled in accordance with the provisions of MIL-STD-105, inspection level II, acceptable quality level (AQL) 1.5 percent defective. For sampling purposes, the lot size shall be the number of units of product in the lot. Examination shall be in accordance with 4.4.1.1.

4.3.6.2 Coiled sheets and strips. Coiled sheets and strips shall be sampled in accordance with the provisions of MIL-STD-105, inspection level II, AQL 2.5 percent defective. For sampling purposes, the lot size shall be the number of coils in the lot. Examination shall be in accordance with 4.4.1.2.

4.4 Inspection.

4.4.1 Visual and dimensional examination.

4.4.1.1 Bars, plates, shapes, and flat sheets and strips. The sampled material of 4.3.6.1 shall be visually examined to determine compliance with 3.7 and 3.8 and dimensionally examined to determine conformance to the applicable dimensional requirements of 3.6. Any unit of product not conforming to any of the requirements of 3.6, 3.7, and 3.8 shall be classed as defective. Acceptance or rejection of the lot shall be based upon an AQL of 1.5 percent defectives.

4.4.1.2 Coiled sheets and strips. The sampled material of 4.3.6.2 shall be visually examined to determine compliance with 3.7 and 3.8 and dimensionally examined to determine conformance to the applicable dimensional requirements of 3.6. A minimum of 10 percent of the area in each coil of the sample shall be examined. Any coil not conforming to any of the requirements of 3.6, 3.7, and 3.8 shall be classed as defective. Acceptance or rejection of the lot shall be based upon an AQL of 2.5 percent defectives.

4.4.2 Packaging, packing, and marking examination. The entire lot shall be visually examined to assure compliance with the packaging, packing, and marking requirements of section 5. Any bundle, lift, or shipping container failing to conform to the requirements of section 5 shall be rejected.

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4.5 Tests.

* 4.5.1 Chemical analysis. The sample(s) of 4.3.1 shall be analyzed in accordance with any of the acceptable procedures specified in method 111 or 112 of FED-STD-151. This analysis shall conform to the compositional requirements for the specified alloy in table I.

* 4.5.2 Hardness. Each specimen of the sample obtained in 4.3.2 shall be prepared and tested in accordance with ASTM E 18. Three Rockwell hardness determinations shall be made on each specimen and the average for each specimen reported. The HRC scale shall be used whenever practicable. In the event that the superficial HRN scales have to be used, table IV of ASTM E 140 shall be used and the results reported as HRC values. Each average hardness value for the specified alloy, form, and condition shall conform to the related requirement in table II.

* 4.5.3 Magnetic effect test. Each specimen of the sample obtained in 4.3.3 shall have its magnetic effect determined in accordance with paragraph 4.4.1 of MIL-M-19595 to assure compliance with the requirement of 3.3 herein.

* 4.5.4 Microstructural examination. Each specimen of 4.3.4 shall be polished, etched (a solution of 5 ml HNO_3:1 ml HF (48%):44 ml distilled water is recommended), and examined at 100 to 200 magnifications on a metallograph to assure compliance with the requirements of 3.4.

4.5.5 Macrostructural examination. Each specimen of 4.3.5 shall be polished to a number 0 metallographic paper finish, deep acid etched (an aqueous solution of HF:HNO_3 or HNO_3 is recommended), and examined to assure compliance with 4.3.5.

4.6 Rejection criteria. Should any sample fail to comply with the specified requirements of this specification or should the lot fail to pass the examination of 4.4.1, the lot shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified.

5.1.1 Level A. The material shall be properly separated by lot number, form, size, and condition and, subsequently, prepared for delivery in accordance with MIL-STD-163.

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5.1.2 Level C. The material shall be properly separated by lot number, form, size, and condition and, subsequently, packaged in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing shall be level A or C, as specified.

5.2.1 Level A. Material packaged as specified in 5.1.1 shall be packed in accordance with MIL-STD-163.

5.2.2 Level C. Material packaged as specified in 5.1.2 shall be packed in accordance with commercial practice to insure carrier acceptance and safe delivery to the destination at the lowest rate. Containers shall conform, as a minimum, to the requirements of the rules and regulations applicable to the mode of transportation selected.

5.3 Marking. In addition to any special markings required by the contract or order and 3.7, the shipments shall be marked in accordance with MIL-STD-163.

6. NOTES

6.1 Intended use. The material covered by this specification is intended for the manufacturing of special purpose tools and equipment where minimum magnetic effects are required. The limits of magnetic effect are far more stringent than those required for general purpose "nonmagnetic" tools.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification

(b) Alloy composition number, form, and condition of material required (see 1.2)

(c) Quantity of material required

(d) Dimensions and, when required, tolerances for the form of material required (see 3.6)

(e) Level of packaging and packing required (see 5.1, 5.2, and 6.2.1)

(f) Special marking when required (see 5.3).

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6.2.1 Criteria for use of proper level of packaging and packing.

6.2.1.1 For Level A. This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

6.2.1.2 For Level C. This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

6.3 Mechanical properties. The nickel-titanium alloys covered by this specification are capable of developing the mechanical properties shown in table III for the various conditions of 1.2. Composition 1 alloy is a nonheat-treatable alloy while compositions 2 and 3 are heat-treatable alloys. The mechanical properties of all 3 compositions can be improved by either hot or cold working.

* 6.4 Recommended conditioning temperature. These nickel-titanium alloys should be heated through at 850° centigrade for hot working, and at about 950° centigrade for the annealing and quench hardening conditions. The mechanical properties for the annealed and quench hardened conditions are dependent upon the rate of cooling from 950° centigrade.

* 6.5 Recommended atmospheric and surface controls. These alloys should be initially produced using arc-casting or vacuum induction melting techniques and may be heat treated in air or inert atmospheres. An oxidizing atmosphere (e.g., air) will cause the formation of adherent oxide coating. This oxide coating may be removed by chemical or mechanical means. To minimize the formation of an oxide coating, an inert atmosphere (e.g., helium or argon) may be employed. For use in nonmagnetic applications, surface oxidation should be avoided or removed as these ferromagnetic surface oxides may increase the magnetic effects of the tests performed in 4.3.3.

6.6 Relationship of volume to magnetic effects. Specimens of these alloys having volumes from 3/4 to 6 cubic inches, when tested in accordance with 4.5.3., produced no measureable effect on magnetometers having a sensitivity of 0.01 milligauss.

* 6.7 Specification information. The Naval Ordnance Laboratory is the design agent for the Naval Ordnance Systems Command having cognizance of this specification which superseded WS 4361B of 28 October 1964.

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Table III

NOMINAL MECHANICAL PROPERTIES OF NICKEL-TITANIUM ALLOYS

Condition	Hardness Rockwell values	Tensile strength (lb/in^2)	Yield strength (lb/in^2)	Total elongation (%)	Impact strength[1] (ft-lb)
		Composition 1			
Cast	86 HRB	80,000	15,000	5	30
Hot worked[2]	20 HRC	120,000	30,000	15	110
		Composition 2			
Hot worked[2]	35 HRC	130,000	-	8	100
Quench hardened in water[2]	40 - 47 HRC	-	-	-	-
		Composition 3			
Cast	52 - 55 HRC	120,000	-	1	19
Hot worked[2]	50 - 53 HRC	170,000	-	1	55
Annealed furnace[2]	27 - 32 HRC	120,000	-	7	50
Quench hardened in water[2]	57 - 62 HRC	150,000	-	1	38

[1]Impact strength was determined using unnotched Charpy specimens.
[2]Material was heated through at 950° centigrade and, subsequently, treated as defined by the specified condition.

* 6.8 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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